

Seongjun Yang

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Research Interests

Standards for Responsible AI Systems

- Developing evaluation frameworks to address the risks and social impacts of generative AI systems, focusing on privacy, bias, and accountability. This involves creating practical benchmarks that align technological progress with regulatory compliance and supporting the ethical integration of AI into high-stakes fields like healthcare and law through transparent, high-confidence decision-making.

Methods Ensuring Trustworthy AI

- Creating methods to ensure users safely achieve their intended outcomes with AI systems, such as LLM agents, under specific goals and constraints. This includes addressing risks such as unintended results, privacy leaks, and algorithmic harm, particularly in high-stakes environments, and proposing empirically and theoretically robust methods to mitigate these vulnerabilities while preserving innovation.

Education

Korea Advanced Institute of Science and Technology

Master's degree, Artificial Intelligence

South Korea

September 2020 - August 2022

Advisor: *Professor. Edward Choi*

Research Area: Federated Learning, Natural Language Processing

Thesis: Towards the Practical Utility of Federated Learning in the Medical Domain

Yonsei University

Bachelor's degree, Computer Science

South Korea

March 2014 - August 2020

Magna cum laude in Dept. of Computer Science

2 years of absence due to obligatory military service (2015 - 2016)

Club Activities

- IronBats (Engineering College Baseball Club)
- Yupalaw (Yonsei University Department of Public Administration Law Society)
- YCC (Yonsei University Computer Club)

Publications

Seongjun Yang*, Gibbeum Lee*, Jaewoong Cho, Dimitris Papailiopoulos, and Kangwook Lee, Predictive Pipelined Decoding: A Compute-Latency Trade-off for Exact LLM Decoding, TMLR 2024

Seongjun Yang*, Hyeonji Hwang*, Daeyoung Kim, Radhika Dua, Jong-Yeup Kim, Eunho Yang, and Edward Choi, Towards the Practical Utility of Federated Learning in the Medical Domain, CHIL 2023

Radhika Dua, **Seongjun Yang**, Yixuan Li, and Edward Choi, Task Agnostic and Post-hoc Unseen Distribution Detection, WACV 2023

Gyubok Lee, Hyeonji Hwang, Seongsu Bae, Yeonsu Kwon, Woncheol Shin, **Seongjun Yang**, Minjoon Seo, Jong-Yeup Kim, and Edward Choi, EHRSQL: A Practical Text-to-SQL Benchmark for Electronic Health Records, NeurIPS 2022 Datasets and Benchmarks

Junu Kim, Kyunghoon Hur, **Seongjun Yang**, and Edward Choi, Universal EHR Federated Learning Framework, In Extended abstract in ML4H 2022

Gyubok Lee*, **Seongjun Yang***, and Edward Choi, Improving Lexically Constrained Neural Machine Translation with Source-Conditioned Masked Span Prediction, ACL 2021 (Short)

Employment History

KRAFTON Inc.	South Korea
NLP Research Engineer	November 2022 – Present
- Investigate methods to reduce the parameter size of LLMs, such as pruning and quantization, to meet GPU requirements without significant performance loss. - Instruct-tune LLMs, such as LLaMA, and developing prompting strategies for in-game applications. - For more details,  PPD,  KORani,  AutoEvalGPT	
NHN Cloud	South Korea
AI Researcher	October 2022 - November 2022
Designed tutorials for benchmarking Korean Language Models.	
Korea Advanced Institute of Science and Technology	South Korea
Graduate Student Researcher	September 2020 - August 2022
Researched Federated Learning and Natural Language Processing under the guidance of Professor Edward Choi	
Teaching Assistant	September 2020 - December 2021
- Served as a teaching assistant for courses titled "Machine Learning for Healthcare", and "Programming for AI" instructed by Professor Edward Choi. - Led tutorials, managed class assignments, and demonstrated key coding skills for up to 100 students at a time.	

Awards and Achievements

Department Prize for Outstanding Student Performance	2019-2, 2020-2
Awarded for achieving grades within the top 3% at Yonsei University.	
Graduation Capstone Design	2019
3rd Award in graduation capstone design program at Yonsei University.	
Industrial Design Competition	2018
- Proposed a system for road damage management 3rd Award (Hosted by South Korean Ministry of Trade, Industry and Energy)	
Grant	
- R.A Scholarship at KAIST - Full Scholarship at Yonsei	
	September 2020 - August 2022 March 2017 - December 2019

Skills & Interests

Technical Skills
Programming Languages: Python, C/C++ Machine Learning Frameworks: PyTorch, TensorFlow Typesetting: L^AT_EX Tools: Git, Linux